


# Permutation & Combination

# PERMUTATION & COMBINATION

Topic is related to counting no. of arrangement / combination.

ABC → ABC, ACB, BAC, BCA, CAB, CBA

Arrangement or Permutation

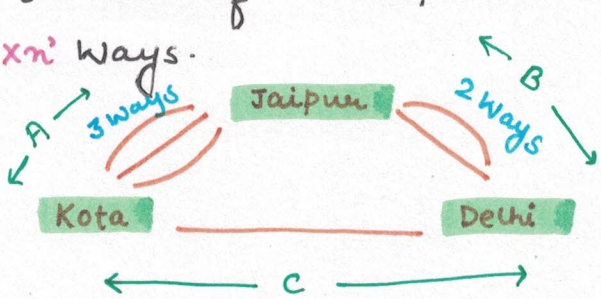
ABCD → Make group containing 2 letter

AB, AC, AD, BC, BD, CD

Combination of making groups

## RULE OF MULTIPLICATION

If A can be done in 'm' ways & B can be done in 'n' ways. Any work 'C' is complete when work 'A' & 'B' is finished. Then rule of multiplication say, 'C' can be completed in 'mxn' ways.



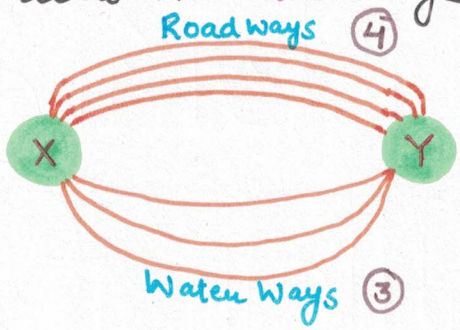

No. of ways in which we can reach Delhi from Kota

$= 3 \times 2 = 6$

## RULE OF ADDITION

If work A can be done in 'm' ways, and work B can be done in 'n' ways and work 'C' is complete either of 'A' & 'B' is done. Then rule of addition says,

C can be done in 'm+n' ways




Total no. of ways to reach Y: 4+3 = 7




1. In how many ways, 4 letters from A, B, C, D can be arrangged?

ABCD
ABDC
ADBC
ADCB
ACBD
ACDB
] ×4 = 6×4 = 24

4 × 3 × 2 × 1 = 4! = 24
↓ ↓ ↓ ↓
$w_1$ $w_2$ $w_3$ $w_4$

2. In how many ways, letters of word 'WORD' can be arranged?

WORD
WODR
WDOR
DWOR
⋮
] 24 ways

3. In how many ways, three letters of word 'WORD' can be arranged?

WOR
ORD
WOD
ORW
⋮
] 24 ways

4×3×2 = 24

4. In how many ways three letters word can be formed by using letter of word ACTORS. (Any letter should not be repeated)

All 5 letter 5 4 3 2 1 = 20×6 = 120

3 letter 6 5 4 = 120

## Arrangement of $r$ objects taken from $n$ ⟹

$$\underset{\uparrow \text{1st place}}{n} \quad \underset{\uparrow \text{2nd}}{n-1} \quad \underset{3^{rd}}{n-2} \quad \cdots\cdots\cdots\cdots \quad \underset{r^{th}\text{ term}}{n-r+1}$$

Total ways $= n \times (n-1) \times \cdots\cdots \times (n-r+1)$

$$= \frac{n(n-1)\times \cdots\cdots (n-r+1)\, r!}{r!}$$

$$= \boxed{\frac{n!}{(n-r)!} = {}^{n}P_{r}}$$ ← Permutation of 'r' things taken from 'n'

AAB, ABA, BAA → 3 ways (letter is repeated, not distinct)

${}^{n}P_{r}$ = no. of permutation (arrangement) of $r$ object taken from $n$ distinct objects.

## Arrangement of $n$ objects of which $r$ objects are same ⟹

AAB → $AA^{*}B$

$AA^{*}B$, $A^{*}AB$ ] Same

$ABA^{*}$, $A^{*}BA$ ] Same

$BAA^{*}$, $BA^{*}A$ ] Same

$= 6$

# of Permutation $= \frac{3!}{2}$

AAAB → AAAB, ABAA, AABA, BAAA ] 4 ways $= \frac{4!}{3!}$

So, permutation of $n$ objects when '$r$' are same is

$$\boxed{\frac{n!}{r!}}$$

# from $n$, $r_1$ → identical, $r_2$ → identical, $r_3$ → identical

$$\text{No. of Permutation} = \frac{n!}{r_1!\, r_2!\, r_3!}$$

Ques: In how many ways the letter of word BANANA can be arranged?

Sol: $\frac{6!}{3!\,2!} = \frac{2\times3\times4\times5\times6}{2\times3\times2} = 10\times6 = \boxed{60}$

Ques: In how many ways, the 5 letters of word BANANA can be arranged?

Sol: ① Letter B, A, N

BANAN $\quad \frac{5!}{2!2!} = 15$

② BANNA $\quad \frac{5!}{2!2!} = 15$

③ AANAN $\quad \frac{5!}{2!3!} = 10$

④ BANAA $\quad \frac{5!}{3!} = 20$

Total ways = 15+15+10+20

= 60

☞ With 4 ltters

| ① | ② | ③ | ④ | ⑤ |
|---|---|---|---|---|
| BANA | AAAN | AANN | BNNA | BAAA |
| $\frac{4!}{2!}$ | $\frac{4!}{3!}$ | $\frac{4!}{2!2!}$ | $\frac{4!}{2!}$ | $\frac{4!}{3!}$ |
| =12 | =4 | =6 | =12 | =4 |

Ques: Using 1, 2, 3, 4, 5, 6, how many 6 digit no's can be created if (a) No.'s are even (b) No's are divisible by 5 (No repetition)

Sol: Method I




⇒ 2×3×4×5×3

⇒ 360

⇒ 5×4×3×2×1

⇒ 120

Method II

All possible no. of numbers = 6×5×4×3×2 = 720

Even = $\frac{720}{2}$ ⇒ 360

Ques: 0, 1, 2, 3, 4, 5, 6

a) even no.

0, 2, 4, 6

| | | | | | |
|---|---|---|---|---|---|
| ↓ | ↓ | ↓ | ↓ | ↓ | ↓ |
| 2 | 3 | 4 | 5 | 6 | 4 |

$= 6 \times 5 \times 4 \times 3 \times 2 \times 1$

$= \boxed{720}$

b) When 0 → unit place

| | | | | | |
|---|---|---|---|---|---|
| | | | | | 0 |
| 2 | 3 | 4 | 5 | 6 | 1 |

$= 720$

When 0 is not on unit place

$5 \times 5 \times 4 \times 3 \times 2 \times 3$

$= 1800$

$1800 + 720 = \boxed{2520}$

## QUESTIONS BASED ON RULE OF MULTIPLICATION:

Ques: In how many ways 4 digit odd no's can be created using 1, 2, 3, 4, 5, 6. (a) rep. not allowed (b) rep. allowed

Sol: Distinct: ③ ④ ⑤ ③ $= 5 \times 3 \times 4 \times 3 = \boxed{180}$

(1, 3, 5)

Repetition Allowed:

⑥ ⑥ ⑥ ③ $= 6 \times 6 \times 6 \times 3 = \boxed{648}$

Ques: How many odd no's can be formed using 1, 2, 3, 4, 5, 6

(a) rep. allowed (b) rep. not allowed

Sol:

⑤ ③ — Two digit → 15

④ ⑤ ③ → $4 \times 5 \times 3 = 60$

③ ④ ⑤ ③ → 180

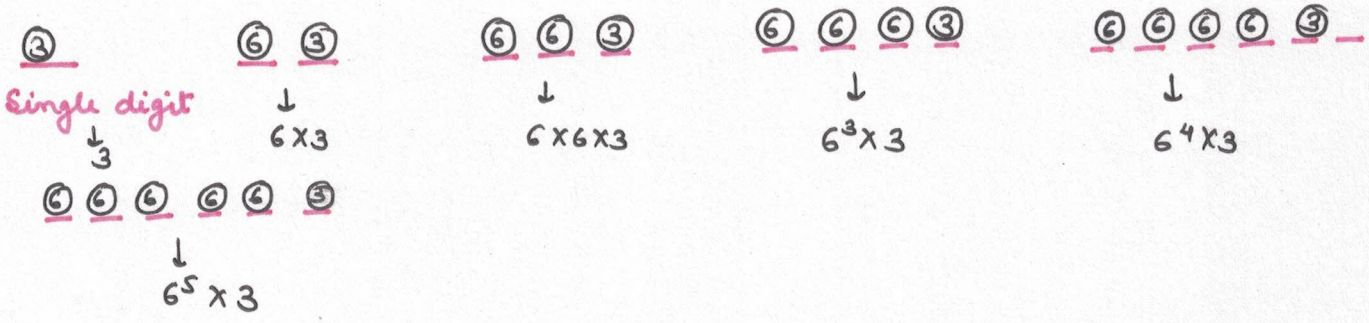
② ③ ④ ⑤ ③ → 360

① ② ③ ④ ⑤ ③ → 360

③ → 3

Total odd no's when rep. not allowed =

$60 + 15 + 180 + 360 + 360 + 3 = \boxed{978}$




Total possible odd no's = $3\left[6+6^2+6^3+6^4+6^5\right]+3$

$= 3\left[\frac{6(1-6^5)}{1-6}\right]+3$

$=$ **27993** → When repetition is allowed

Ques: How many numbers of 4 digit can be formed using 1, 2, 3, 4, 5., so that no's are:

a) multiple of 5

b) multiple of 2

c) multiple of 3

PRACTISE

## POINTS TO REMEMBER

- Permutation of $n$ distinct objects = $n!$
- Permutation of '$r$' objects taken from $n$ distinct object is:-

$$^nP_r = \frac{n!}{(n-r)!}$$

- Permutation of '$n$' objects when $r$ of them are identical :-

$$\frac{n!}{r!}$$

- Permutation of '$n$' objects where $r_1$ of them is one kind (identical) and $r_2$ of them are other kind is:-

$$\frac{n!}{r_1!\, r_2!}$$

Ques 1: In how many ways the letter of word 'READ' can be arranged

Sol: $4! = 4\times3\times2\times1 =$ **24**

Ques 2: How many words can be formed using all the letters of word 'JAPANESE'?

Sol: $\frac{8!}{2!\,2!} = \frac{2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 \cdot 7 \cdot 8}{2 \cdot 2} = \boxed{10080}$

Ques 3: How many 5 digit numbers can be formed using 1, 2, 3, 4, 5, 6, 7

Sol: $^7P_5 = \frac{7!}{2!} = 3 \times 4 \times 5 \times 6 \times 7 = \boxed{2520}$

Ques 4: How many 4 digit no's can be formed by using 0, 1, 2, 3, 4, 5?

Sol:- I) ⑤ ⑤ ④ ③ $0 \neq$ first place

↓

300

Alternatively, II) _ _ _ _ $^6P_4 - {}^5P_3 = \frac{6!}{2!} - \frac{5!}{2!} = 360 - 60$

$= \boxed{300}$

Where, $^6P_4 \rightarrow$ all 4 digit including '0' at thousand place i.e. 3 digit no + 4 digit no.

$^5P_3 \rightarrow$ all 3 digit no. from digits 1, 2, 3, 4, 5

All 4 digit no's - $\boxed{^6P_4 - {}^5P_3}$

Ques5: $^{56}P_{r+6} : {}^{54}P_{r+3} = 30800 : 1$, Find $^rP_2 = ?$

Sol: $\frac{56!}{(56-r-6)!} \times \frac{(54-r-3)!}{54!} = \frac{30800}{1}$

$\Rightarrow \frac{55 \times 56}{(50-r)!} \cdot (51-r)! = \frac{30800}{1}$

$\Rightarrow 55 \cdot 56\,[51-r] = 30800$

$\Rightarrow 51-r = 10$

$\boxed{r = 41}$

therefore, $^{41}P_2 = \frac{41!}{39!} = 40 \times 41 =$ **1640**

Ques 6: $^5P_r = {}^6P_{r-1}$, then show that $r = 4$

Sol: $$\frac{5!}{(5-r)!} = \frac{6!}{(6-r+1)!}$$

$$\Rightarrow \quad \frac{1}{(5-r)!} = \frac{6}{(7-r)!}$$

$$\Rightarrow \quad (6-r)(7-r) = 6$$

$$\Rightarrow \quad 42 - 6r - 7r + r^2 = 6$$

$$\Rightarrow \quad r^2 - 13r + 36 = 0$$

$$\Rightarrow \quad r^2 - 9r - 4r + 36 = 0$$

$$\Rightarrow \quad r = 9$$

**$r = 4$**

$r \neq 9$
$r \neq 5$ or $6$

Ques 7: 7 letter words using 'ARIHANT'

Sol: A → 2 times

$$\frac{7!}{2!} = 3 \cdot 4 \cdot 5 \cdot 6 \cdot 7 =$$ **2520**

Ques 8: Prove that $^nP_r = {}^{n-1}P_r + r \cdot {}^{n-1}P_{r-1}$

Sol: $$\frac{(n-1)!}{(n-1-r)!} + r \cdot \frac{(n-1)!}{(n-1-r+1)!}$$

$$\Rightarrow \quad \frac{(n-1)!\,(n-r) + r\,(n-1)!}{(n-r)!}$$

$$\Rightarrow \quad \frac{(n-1)!\,[n]}{(n-r)!}$$

$$\Rightarrow \quad \frac{n!}{(n-r)!} = \boxed{^nP_r}$$




Ques 9: 'MATHEMATICS' – Total arrangements?

Sol: $$\frac{11!}{2!\,2!\,2!}$$ } Total arrangements

Ques 10: 'LUCKNOW'

Sol: i) All letters = $^7P_7$ → 7!

ii) Letter begin with L = L _ _ _ _ _ _ → 6!

iii) L- first, W- Last = L _ _ _ _ _ W → 5!

iv) Vowels = O and U = 2



→ 6!2!

# COMBINATION

Selection → Making groups

$^nC_r$ → Selection of r objects from 'n' distinct objects

Example: 3 persons - A, B, C → Make group of 2

$^3C_2$ → AB, BC, CA

★ Permute 2 people out of 3:

$^3P_2$ → AB, BA, CB, BC, CA, AC

★ Permutation of 2 out of 3: Making group of 2 & then permuting them

$^3P_2 \rightarrow {}^3C_2 \times 2!$

$^3C_2 = \frac{^3P_2}{2!}$

★ Permutation of 'r' objects out of 'n': Making group of r and then arranging each group

$^nP_r = {}^nC_r \times r!$

$^nC_r = \frac{^nP_r}{r!}$

$$^nC_r = \frac{n!}{(n-r)!\,r!}$$

} No. of making groups of r objects from n

Ques: Find no. of ways in which a badminton doubles team is formed from 5 players.

Sol: ${}^5C_2 = \frac{5!}{3!\,2!} = \boxed{10}$

Ques: How many $\Delta$'s can be formed using 10 points. (None of the 2 points are collinear)

Sol: ${}^{10}C_3 = \frac{10!}{7!\,3!} = \boxed{120}$

Ques: Find the max. possible straight line that can be drawn using 10 points.

Sol: 10 points → non- collinear

${}^{10}C_2 = \frac{10!}{8!\,2!} = \boxed{45}$ Max. lines

* Min. no. of lines = 1 [When all 10 pts. are collinear]

Ques: Find the no. of possible handshakes in a room containing 20 people [No 2 people shake hands more than once]

Sol: ${}^{20}C_2 = \frac{20!}{18!\,2!} = \boxed{190}$

Ques: ${}^{2n}C_3 : {}^nC_2 = 12:1$, find $n$?

Sol: $\frac{(2n)!}{(2n-3)!\,3!} \times \frac{(n-2)!\,2!}{n!} = \frac{12}{1}$

$\Rightarrow \frac{(2n-2)(2n-1)\,2n}{(n-1)\,n} = 36$

$\Rightarrow 4n^2 - 2n - 4n + 2 = 18n - 18$

$\Rightarrow 4n^2 - 6n - 18n + 20 = 0$

$\Rightarrow 4n^2 - 24n + 20 = 0$

$\Rightarrow n^2 - 6n + 5 = 0$

$\Rightarrow n^2 - 5n - n + 5 = 0$

$\Rightarrow n = 5 \quad n = -1$ $\boxed{N=5}$

Ques: ${}^{25}C_{n+5} = {}^{25}C_{2n-1}$, Find $n$?

Sol: $$\frac{25!}{(25-n-5)!\,(n+5)!} = \frac{25!}{(25-2n+1)!\,(2n-1)!}$$

$\Rightarrow$ $(26-2n)!\,(2n-1)! = (20-n)!\,(n+5)!$

$\Rightarrow$ $[2(13-n)]!\,(2n-1)! = (20-n)!\,(n+5)!$

$\Rightarrow$ $n+5 = 2n-1$

$$\boxed{n = 6}$$

OR $n+5+2n-1 = 25$

$\Rightarrow$ $3n+4 = 25$

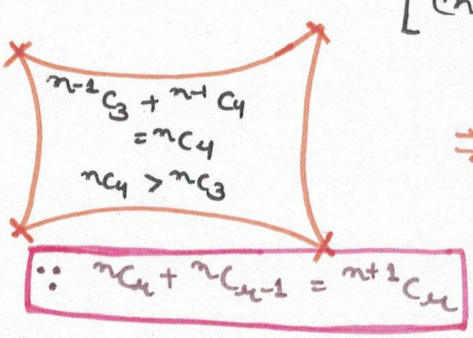
$\Rightarrow$ $$\boxed{n = 7}$$

Ques: ${}^{n-1}C_3 + {}^{n-1}C_4 > {}^{n}C_3$ if $n>7$

Sol: $$\frac{(n-1)!}{3!\,(n-1-3)!} + \frac{(n-1)!}{4!\,(n-1-4)!} > \frac{n!}{(n-3)!\,3!}$$

$\Rightarrow$ $$\frac{(n-1)!\,(n-4)\,4! + (n-1)!\,3!}{(n-5)!\,3!\,4!} > \frac{n!}{(n-3)!\,3!}$$

$\Rightarrow$ $$(n-1)!\left[\frac{24n-96+6n-6}{(n-5)!\,3!\,4!}\right] > \frac{n!\,(n-1)}{(n-3)!\,3!}$$




$\Rightarrow$ $$\frac{n!}{(n-4)!\,4!} > \frac{n!}{(n-3)!\,3!}$$

$\Rightarrow$ $n-3 > 4$

$$\boxed{n > 7}$$

- Selection of $r$ out of $n$ distinct objects = ${}^{n}C_r$
- Selection of atleast one object from $n$ distinct object = ${}^{n}C_1 + {}^{n}C_2 + {}^{n}C_3 - - - - - {}^{n}C_n = 2^n - 1$
- Selection from identical objects - Total possible selection from $n$ identical objects = $(n+1)$
- $n \rightarrow$ different $\quad p \rightarrow$ identical (one kind) $\quad q \rightarrow$ unidentical (another kind)

no. of selection of atleast one object: $(p+1)(q+1)2^n - 1$

Ques: The total no. of selections (taking at least one) of fruit which can be made from 3 Bananas, 4 apples and 2 oranges.

Sol:

3 bananas → $(3+1) = 4$

4 apples → $(4+1) = 5$

2 oranges → $(2+1) = 3$

Total no. of selections $= 4 \times 5 \times 3 - 1$

$= 60 - 1$

$= \boxed{59}$

Ques: DEGREE

Sol: ${}^{6}C_{4} = \frac{6!}{4!\,2!} \times 3! = \boxed{\frac{6!}{8}}$

Ques: Show that the no. of ways of selecting n-objects out of 3n objects, n of which are alike and rest different is ${}^{2n}C_{n} + {}^{2n}C_{n-1} + \dots + {}^{2n}C_{0}$

Sol: $n \rightarrow$ alike $\qquad 2n \rightarrow$ different

${}^{2n}C_{n} + {}^{2n}C_{n-1} + \dots + {}^{2n}C_{0}$

Identical ↑ ; ↑ Non-identical

$1 \times {}^{2n}C_{n} + 1 \times {}^{2n}C_{n-1} + 1 \times {}^{2n}C_{n-2} + \dots + 1 \times {}^{2n}C_{0}$

from Identical ↑ ; ↑ from non-identical

| n- alike | no. of ways | 2n different object | no. of ways |
|---|---|---|---|
| 0 | 1 | n | ${}^{2n}C_{n}$ |
| 1 | 1 | n-1 | ${}^{2n}C_{n-1}$ |
| 2 | 1 | n-2 | ${}^{2n}C_{n-2}$ |
| ⋮ | ⋮ | ⋮ | ⋮ |
| n | 1 | 0 | ${}^{2n}C_{0}$ |

Ques: 1, 2, 3, 4, 2, 1, odd digits occupy odd places.

Sol: 3 → alike (2)

2 → alike (2)

1 → alike (1)

3 → (2+1) = 3

2 → (2+1) = 3

1 → (1+1) = 2

1 2 3 2 1 4 3

3 2 1 2 3 4 1

Total ways $= 3 \times 3 \times 2$

$= \boxed{18}$

No. of ways of filling odd places by odd digits

$= \frac{4!}{2!2!} = 6$

No. of ways of filling even places by even digits

$= \frac{3!}{2!} = 3$

Total no. of ways = **18**

## CIRCULAR PERMUTATION

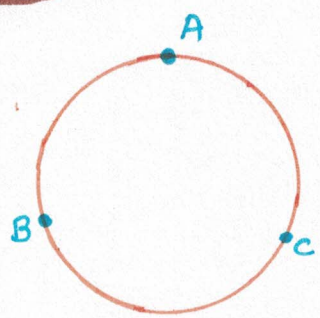


Arranging n distinct objects in a circle has $(n-1)!$ ways

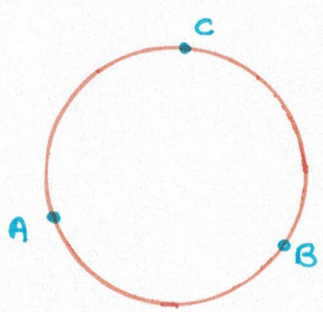
- - - - - - - - - } Linear arrangement

$n!$ Ways

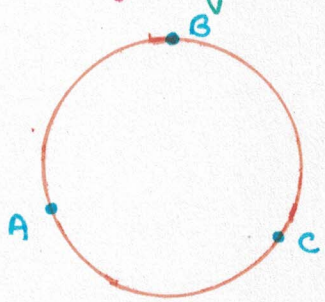
Example:




A B C




C A B




B A C

No. of circular permutation = x

$x \times 3$ = No. of permutation of 3

$x \times 3 = 3!$

$x = \frac{3!}{3} = 2!$

**$x = (3-1)!$**

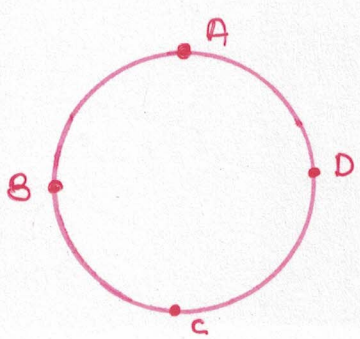
Example:

A B C D

C D A B

B C D A

D A B C



Total Way of circular permutation = $x$

Linear arrangement = $4x$

$4x = 4!$

$x = \frac{4!}{4} = 3! =$ **$(4-1)!$**

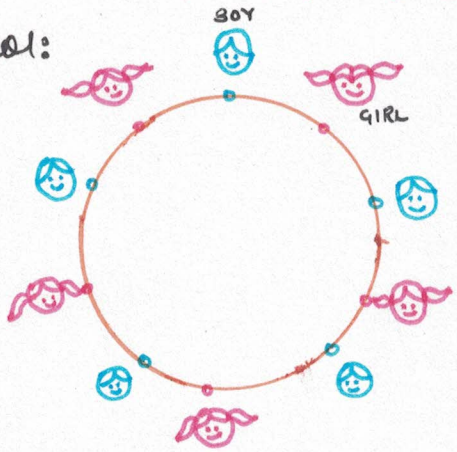
Another method of understanding this is, place one person at a position. Now $(n-1)$ places are left unique. 1st person – 1 choice 2nd person – $(n-1)$ choice 3rd person – $(n-2)$ choice

Total no. of choice:

$1(n-1)(n-2)(n-3)\text{-}\text{-}\text{-}\text{-}\text{-} = (n-1)!$

Ques: In how many different ways can 5 boys & 5 girls form a circle such that the boys and girls are alternate?

Sol:




Ways of arrangement of boys = 4!

Ways of arrangement of girls = 5!

Total = 4!5!

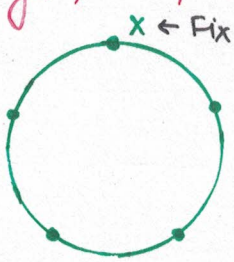
= **2880**

Ques: Find the no. of seating arrangements for 5 person when chairs are around circular table.

Sol: $(5-1)! = 4! =$ **24**

Ques: There are 10 chairs in a circle. In how many ways, 5 persons can be seated on the chairs.

Sol:




Places for remaining 4 persons = 9

Arrangements = ${}^{9}P_4$

$= \frac{9!}{5!} = 72 \times 42$

= **3024**

# CIRCULAR PERMUTATION

1 Permutation of $n$ objects (distinct) in a circle:

$$(n-1)!$$

2 Permutation of $n$ objects taken $r$ at a time in a circle:

$${}^{n}P_{r}$$

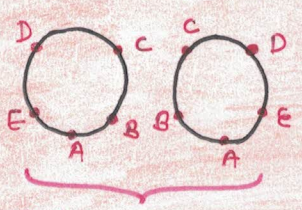
Ques: Find the no. of ways in which 4 people selected from 5 can be seated around a round table

Sol: $$\frac{{}^{5}P_{4}}{4} = \frac{5!}{(5-4)!} \times \frac{1}{4} = \frac{120}{4} = \boxed{30}$$

3 Arrangements of $n$ beads in necklace

$$\frac{(n-1)!}{2}$$

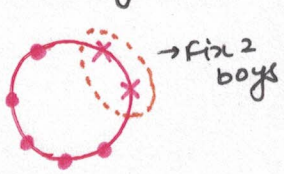
$(n-1)!$ is the no. of circular arrangements. We will always get pairs of circular arrangement which will give one necklace.



Both are same in necklace

PRACTISE PROBLEM

1 (i) 7 boys → 2 boys should sit together

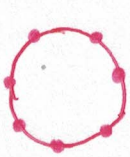



$= (6-1)!\,2!$

$= 5!\,2! = 120 \times 2$

$= \boxed{240}$

(ii) 7 boys



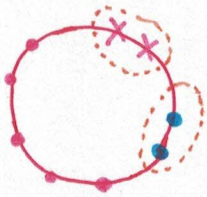
Total Ways $= 6! = 120 \times 6 = 720$

Separated $= 720 - 240$

$= \boxed{480}$

2. 5 boys, 4 girls

2 girls → Not together  2 boys → not together



$(7-1)!\ 2!\ 2!$

$= 6! \times 4 = 720 \times 4 = \boxed{2880}$

Total ways $= 8! = 720 \times 7 \times 8 = \boxed{40320}$

3. 5 boys, 4 girls

Two boys → together

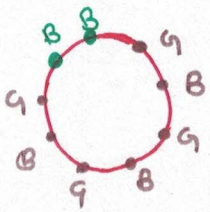


$= 4!\,3!\,2!$

$= 24 \times 6 \times 2$

$= 144 \times 2$

$= \boxed{288}$




4. 13 chairs, 10 students, 1 teacher } 11 people




3 chairs → fix

2 people - fix

Linear way { 13 chairs, 7 people } $= {}^{13}P_7 = \frac{13!}{6!}$

Circular $= \frac{13!}{6!\,7} = 8 \times 9 \times 10 \times 11 \times 12 \times 13$

No. of Ways $= \frac{{}^{n}P_r}{r}$

Selections of 2 students from 10 for sitting on chairs beside Matthew $= {}^{10}C_2$

No. of ways in which 2 students can be seated beside Matthew $= {}^{10}C_2 \times 2!$

Arrangement of remaining students $= {}^{10}P_8$

**Total Arrangement $= {}^{10}C_2 \times 2! \times {}^{10}P_8$**

# DIVISION INTO GROUPS/PARTITION INTO GROUPS

* Making group is different from partition

Example: A, B, C divide into (partition into) 2 groups one containing 1 object and other 2 objects.

A | BC  B | AC  C | AB → 3 ways

Example: A, B, C, D, E divide into 2 parts, one containing 2 objects and other 3 objects.

AB | CDE

BC | ADE

⋮

$^5C_2 = \frac{5!}{2!\,3!}$

a) Division of $n$ object into 2 groups of size $r_1$ and $r_2$

$$^nC_{r_1} = \frac{n!}{r_1!\,(n-r_1)!} = \boxed{\frac{n!}{r_1!\,r_2!}}$$

b) Division of $n$ object into 3 groups containing $r_1$, $r_2$, $r_3$ objects. $(r_1 \neq r_2 \neq r_3)$

$$= {}^nC_{r_1}\ {}^{n-r_1}C_{r_2}\ {}^{r_3}C_{r_3}$$

$$= \frac{n!}{r!\,(n-r_1)!} \times \frac{(n-r_1)!}{r_2!\,(n-r_1-r_2)!}$$

$$= \boxed{\frac{n!}{r_1!\,r_2!\,r_3!}}$$

Ques: Find the no. of ways of dividing 13 objects in 3 groups containing 8, 2 and 3 objects

Sol: $\frac{13!}{8!\,2!\,3!}$

Ques: Divide A, B, C, D into 2 group each containing 2.

Sol: AB | CD  AD | BC  AC | BD

3 groups

$$\frac{4!}{2!\,2!\,2!} = \frac{4\times3\times2}{2\times2\times2} = 3 \text{ groups}$$

a) Making group is different from partition

ABCD

Groups
AB, BC, CD,
AD, AC, BD
$^4C_2$

Division into groups / partition
ab | cd, ac | db, ad | bc,
cd | ab, bd | ac, bc | ad
Total → 3 partitions

b) When groups are of same size, When n objects are divided into 2 groups of same size (r)

$$\frac{n!}{r!\,r!}\times\frac{1}{2!} \qquad (2r=n)$$

When n objects are divided into 3 groups of equal size (r)

$$\frac{n!}{r!\,r!\,r!}\times\frac{1}{3!}$$

c) When n object are partitioned as follows in 7 groups

→ 3 groups of size $r_1$ → 2 groups of size $r_2$

→ 1 group of size $r_3$ → 1 group of size $r_4$

$$\frac{n!}{(r_1!)^3\times3!\;(r_2!)^2\times2!\;r_3!\;r_4!}$$

Ques: Find the no. of ways of partition of 13 objects in 5 groups of size → 2, 2, 4, 4, 1

Sol:

$$\frac{13!}{2!\,2!\,2!\;4!\;4!\;2!\;1!}$$

$$= \frac{5\times6\times7\times8\times9\times10\times11\times12\times13}{2\times2\times2\times2\times3\times4\times2}$$

$$= 130\times63\times5\times3\times11$$

$$= 10395\times130$$

## DISTRIBUTION OF OBJECT AMONG PEOPLE

| a, b, c | | P, Q |
|---|---|---|
| Object | | Person |
| a / bc | ⟶ | 2 |
| b / ac | ⟶ | 2 |
| c / ad | ⟶ | 2 |
| $\frac{3!}{1!2!} = 3$ | | $3\times2! = 6$ |

- Distribution of $n$ object among $r$ people
  1. Partition of the object in groups
  2. Distributi among the people

Ques: Find the no. of ways of dividing 4 objects among 3 persons in such a way that all get atleast 1 object.

Sol: 2 1 1 } 1 way

1) $\frac{4!}{2!1!2!} = 6$

2) $6\times3! = 6\times6 = \boxed{36}$

N Objects (among 7 person)

→ 3 groups of $x_1$ size → 2 groups of $x_2$ size

→ 1 group of $x_3$ size → 1 group of $x_4$ size

No. of ways of dividing n objects:

$$\frac{n!}{(x_1)^3 3! (x_2!)^2 2! (x_3)! (x_4)!} \times 7!$$

Ques: Find the no. of ways of dividing 5 objects in 3 person?

Sol:



3 1 1

$$= \frac{5!}{3!\,1!\,1!}$$

$$= \frac{4 \times 5}{2}$$

$$= 10 \times 3!$$

$$= 60$$



2 2 1

$$= \frac{5!}{2!\,2!\,1!}$$

$$= \frac{3 \times 4 \times 5}{2 \times 2}$$

$$= 15 \times 3!$$

$$= 90$$

Total = 60 + 90 = 150 ways

Ques: Find the no. of ways of division of 7 objects in 3 groups each containing atleast 2 objects, when:-

a) objects are identical b) distinct

Sol:-

| Group A | Group B | Group C | |
|---|---|---|---|
| 2 | 2 | 3 | → only 1 group choice |

a) If all are identical there is only 1 Way

b) If distinct, $\frac{7!}{2!2!3!2!} = \frac{4 \times 5 \times 6 \times 7}{2 \times 2 \times 2}$

$= 35 \times 3 =$ 105 Ways

Ques: Find the no. of ways of distributing 7 objects among 3 friends such that all get atleast 1 and 2 get same no. of objects.

Sol:

a) 2 2 3

1 3 3

1 1 5

No. of group × (3) → No. of person

= 3×3

= 9

b)

2 2 3 → $\frac{7!}{2!2!3!} \times \frac{3!}{2!} = 630$

1 3 3 → $\frac{7!}{1!3!3!} \times \frac{3!}{2!} = 420$

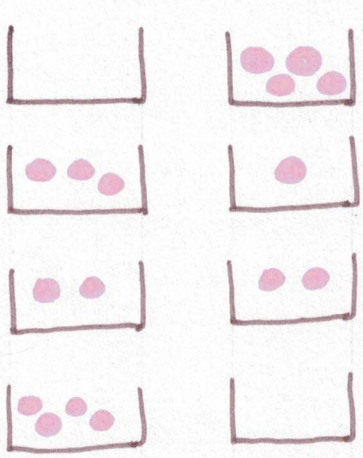
1 1 5 → $\frac{7!}{1!1!5!} \times \frac{3!}{2!} = 126$

Total No. of ways = 630 + 420 + 126 = 1176

## DISTRIBUTION OF IDENTICAL OBJECTS

(a) 4 identical object, distributi in 2 boxes. (Empty boxes are allowed)

| A | B |
|---|---|
| 0 | 4 |
| 3 | 1 |
| 2 | 2 |
| 4 | 0 |



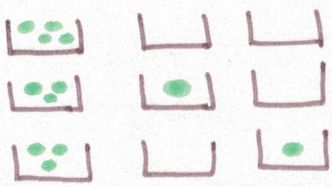
No. of ways of distribution of 4 identical balls is equal to arrangements of 1 marker

and 4 identical balls:

$$= \frac{5!}{4!} = {}^5C_4$$

(b) 4 identical object, distribute into 3 boxes (empty boxes are allowed)

| A | B | C |
|---|---|---|
| 4 | 0 | 0 |
| 3 | 1 | 0 |
| 3 | 0 | 1 |
| ⋮ | ⋮ | ⋮ |



Equivalent to arrangement of 4 identical balls and 2 markers (identical)

$$= \frac{6!}{4!2!} = {}^6C_2$$

(c) n identical objects distributed among r boxes (empty boxes are allowed) ⇒ (r-1) markers are needed

Total objects = n identical balls + (r-1) identical markers

**Ways of arrangement = $\frac{(n+r-1)!}{n!(r-1)!} = {}^{n+r-1}C_{r-1}$**

# When empty boxes are not allowed

↳ n objects (identical)

↳ r boxes

Finding possible no. of distribution is equivalent to finding no. of ways of arrangement of (r-1) markers between n balls, so that no 2 markers come together.

(n-1) places for (r-1) markers → 1 way of n balls (arranged)

**Distribution of n identical object in 1 way of n balls (arranged) boxes (empty not allowed is $= {}^{n-1}C_{r-1}$**

Q Find the no. of ways in which 20 identical objects are distributed in 4 boxes

i) Empty boxes are allowed

ii) Empty boxes are not allowed

Sol: a) ${}^{n+r-1}C_{r-1} = {}^{24-1}C_3$

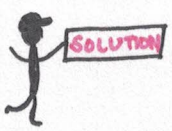
$= \frac{23!}{20!3!} = \frac{21 \times 22 \times 23}{2 \times 3}$

$= \boxed{1771}$

b) ${}^{n-1}C_{r-1} = {}^{19}C_3$

$= \frac{19!}{16!3!} = \frac{17 \times 18 \times 19}{3 \times 2}$

$= \boxed{969}$




If question was asked like this -

Find no. of solution of $x_1 + x_2 + x_3 + x_4 = 20$

Where $x_i$ are non negative integar.

Then also answer will be same.

The no. of sol. is equal to the no. of ways of distributing 20 identical objects in 4 boxes

$\Rightarrow$ No. of solution of $x_1 + x_2 + x_3 + x_4 = 20$ ($x_i \neq$ -ve integar) is -

${}^{20+3}C_3$

No. of solution of $x_1 + x_2 + x_3 + \dots + x_r = n$, where $x_i$'s $\geq 0$ and integer -

${}^{n+r-1}C_{r-1}$

Ques: Find the no. of solutions of equation of, $x_1+x_2+x_3+x_4+x_5 = 10$, where $x_i$'s are non-negative integer

Sol: $^{10+5-1}C_{5-1} = {}^{14}C_4$

$$= \frac{14!}{10!4!}$$

$$= \frac{11\times12\times13\times14}{2\times3\times4}$$

$$= 1001$$

Ques: Find the no. of solution of $x_1+x_2+x_3+x_4 = 12$, where $x_1 \geq 0$, $x_2 \geq 0$, $x_3 \geq 2$, $x_4 \geq 0$

Sol: If all are non-negative, then solution -

$$^{12+3}C_3 = \frac{15!}{12!3!} = 13\times14\times15$$

Put $t_3 = x_3 - 2$

$$x_1 + x_2 + t_3 + 2 + x_4 = 12$$

$$x_1 + x_2 + t_3 + x_4 = 10$$

No. of sol. $= {}^{10+4-1}C_3 = {}^{13}C_3$

$$= \frac{10!}{3!}$$

$$= 4\times5\times6\times7\times8\times9\times10$$

$$= 604800$$

## FINDING NO. OF SOLUTION OF INEQUALITY [ELSE OF DUMMY VARIABLE]

Example: $x_1+x_2+x_3 \leq 15$ where $x_1, x_2, x_3 \geq 0$ and integer

Add new variable $z$, where $0 \leq z < 15$

$$x_1 + x_2 + x_3 + z = 15$$

No. of solution = $^{18}C_3$

Ques. Find the no. of solution :

i) $x_1 + x_2 + x_3 + x_4 = 15$ where, $x_1 \geq 3$, $x_2 \geq -2$, $x_3 \geq 3$ $x_4 \geq 0$

$t_1 + t_2 + t_3 + t_4 = 15 - 4 = 11$

$^{11+4-1}C_3 = {}^{14}C_3 = \frac{14!}{11.3!}$

$= 374$

ii) $x_1 + x_2 + x_3 + x_4 + x_5 = 20$ where $x_i s \geq 0$

$x_1 + x_2 + x_3 + x_4 + x_5 + z = 19$

$^{19+4-1}C_5 = {}^{22}C_5$

iii) $x_1 + x_2 + x_3 + x_4 < 20$ where $x_i s \geq 0$

$x_1 + x_2 + x_3 + x_4 + z = 19$

$^{19+4-1}C_4 = {}^{22}C_4$

iv) $x_1 + x_2 + x_3 < 10$ where $x_1 \geq 2$, $x_2 > 1$, $x_3 \geq 0$, $x_2 \geq 2$

$t_1 + t_2 + t_3 + 4 < 10$

$t_1 + t_2 + t_3 + z = 5$

$^{5+3}C_3 = {}^{8}C_3$

## APPLICATION OF MULTINOMIAL THEOREM

$(x^0 + x^1 + x^2)$, $(x^0 + x^1)$

constant term = 1 coefficient of $x^1 = 1 + 1 = 2$

coefficient of $x^2 = 1 + 1 = 2$ coefficient of $x^3 = 1$

→ 0, 0 x

No. of selection of zero object = 1

No. of ways of selection of 1 object = 2

" " " 2 objects = 2

" " " 3 object = 1

→ $(1+x^1+x^2+x^3)(1+x^1+x^2+x^3)$

0 0 0 x x x

Select 4 objects –

0 x x x → coefficient of $x^1 \cdot x^3$

0 0 x x → coefficient of $x^2 \cdot x^2$

0 0 0 x → coefficient of $x^3 \cdot x$

3 ways

→ 2 groups of identical objects

$r_1$ → ONE Kind

$r_2$ → Another Kind

No. of ways of selection of s object is equal to coefficient of $x^s$ in $(1+x+x^2+\dots x^{r_1}) \times (1+x+x^2+\dots x^{r_2})$

Ques: Find the no. of ways in which 5 objects can be selected from 7 white and 10 black balls

Sol:

White: 0 0 0 0 0 0 0

$(1+x^2+x^3+x^4+x^5+x^6+x^7+x^8)$

Black: 0 0 0 0 0 0 0 0 0 0 0

$(1+x^2+x^3+x^4+x^5+x^6+x^7+x^8+x^9+x^{10}+x^{11})$

$x^5 = 1+1+1+1+1+1$

$x^5 = 6$

METHOD-2

The no. of selection 5 objects is equal to the coefficient of $x^5$ in $(1+x+x^2+\dots x^7)(1+x+x^2+\dots x^{10})$

$= \left(\frac{1-x^8}{1-x}\right)\left(\frac{1-x^{12}}{1-x}\right)$

$= (1-x^8)(1-x^{11})(1-x)^{-2}$

$= (1-x^8-x^{11}+x^{21})(1-x)^{-2}$

Equivalent to coefficient of $x^5$ in $(1-x)^{-2}$

$= 1 + 2x + \frac{(-2)(-2-1)}{2!}(-x)^2 + \dots\dots$

$= 1 + 2x + {}^3C_2 x^2 + {}^4C_3 x^3 + {}^5C_4 x^4 + {}^6C_5 x^5 + \dots$

Coefficient of $x^5 = {}^6C_5 = \frac{6!}{5!1!} = 6$

Ques: $(1+x^1+x^2+x^3+x^4+x^5+x^6)(1+x^2+x^3+\dots x^7)(1+x^2+x^3+\dots x^8)$.

Find coefficient of $x^{10}$.

sol: $\left(\frac{1+x^7}{1-x}\right)\left(\frac{1+x^8}{1-x}\right)\left(\frac{1+x^9}{1-x}\right)$

$= (1+x^8+x^7+x^{15})(1+x^9)(1-x)^{-3}$

$= x^9 + x^{17} + x^{11} + x^{22} + 1 + x^8 + x^7 + x^{15}$

$= 1 + {}^3C_1 x + {}^4C_2 x^2 + {}^5C_3 x^3 + {}^5C_4 x^4 + \dots {}^{11}C_{10} x^{10}$

Coefficient of $x^{10} = {}^{11}C_{10} + {}^4C_2 + {}^3C_1 + {}^5C_3 = {}^7C_3 = 35$

Ques: Coefficient of $x^5$ - $(1+x+x^2+\dots x^{10})^3$

sol: $\left(\frac{1+x^{11}}{1-x}\right)^3 = (1+x^{11})^3 (1-x)^{-3}$

Coeff. of $x^5$ in $(1-x)^{-3}$

$= 1 + 3x + \frac{(3)(2)}{2!} + \frac{(2)(1)}{2!} + \dots$

$= 1 + {}^3C_1 x + {}^4C_2 x^2 + {}^5C_3 x^3 + {}^6C_4 x^4 + \dots\dots$

Coefficient of $x^5 = {}^7C_5 = \frac{7!}{5!2!} = 21$

1. Number of ways of selection of atleast one object from each then multinomial will be like-

$(x+x^2+x^3+\dots x^{n_1})(x+x^2+\dots x^{n_2})(x+x^2+x^3+\dots x^{n_3})$

group 1 contains no. of object $= n_1$

—"— 2 —— "—— "—— $= n_2$

—"— 3 —— "—— "—— $= n_3$

2. The number of ways of selection of 's' object from 3 groups of R, G, and B balls when the condition is -

$\alpha_1 \le$ # of selected R balls $\le \alpha_2$

$\beta_1 \le$ # of selected G balls $\le \beta_2$

$\gamma_1 \le$ # of selected B balls $\le \gamma_2$

is coefficient of $x^s$ in -

$(x^{\alpha_1} + x^{\alpha_1+1} + x^{\alpha_1+2} + \dots x^{\alpha_2})(x^{\beta_1} + x^{\beta_1+1} + x^{\beta_1+2} + \dots x^{\beta_2})$
$(x^{\gamma_1} + x^{\gamma_1+1} + x^{\gamma_1+2} + \dots x^{\gamma_2})$

## NUMBER OF DIVISORS

$N = P_1^{\alpha_1} P_2^{\alpha_2} \dots P_K^{\alpha_K}$ (Prime factorisation)

Eg:
$60 = 2^2 \cdot 3^1 \cdot 5^1$

No. of divisors = ?

Ways of selection:

$P_1 \longrightarrow \alpha_1 + 1$

$P_2 \longrightarrow \alpha_2 + 1$

$\vdots$

$P_K \longrightarrow \alpha_K + 1$

- No. of divisors $= (\alpha_1+1)(\alpha_2+1)____(\alpha_K+1)$
- No. of proper divisors $= (\alpha_1+1)(\alpha_2+1)+___(\alpha_K+1)-2$

Ques: Find the no. of divisors of 120

Sol: $120 = 2^3 \times 3^1 \times 5^1$

$4 \times 4 =$ **16**

Ques: Find the no. of even divisors of 120.

Sol: Method 1:

$120 = 2^3 \times 3^1 \times 5^1$

Ways of selecting $2 \rightarrow 3$

(atleast one 2) even divisors $= 3 \times 2 \times 2$

Ways $=$ **12**

Method 2: odd divisors $= (1+1)(1+1) = 4$

Even divisors $= 16-4 =$ **12**

- Sum of all divisors $= N = P_1^{\alpha_1} P_2^{\alpha_2} ___ P_K^{\alpha_K}$
- Sum of divisors $= (1+P_1+P_1^2+___P_1^{\alpha_1})(1+P_2+P_2^2+__P_2^{\alpha_2})+$
$_____+(1+P_K+P_K^2+___P_K^{\alpha_K})$

$$= \left(\frac{P_1^{\alpha_1+1}-1}{P_1-1}\right)\left(\frac{P_2^{\alpha_2+1}-1}{P_2-1}\right)_____\left(\frac{P_K^{\alpha_K+1}-1}{P_K-1}\right)$$

Ques: Sum of all divisors of 120

Sol: $120 = 2^3 \times 3^1 \times 5^1$

$$= \left(\frac{2^4-1}{2-1}\right)\left(\frac{3^2-1}{3-1}\right)\left(\frac{5^2-1}{5-1}\right)$$

$$= \frac{15}{1} \cdot \frac{8}{2} \cdot \frac{24}{4} = 120 \times 3 = \boxed{360}$$

Ques: Find the sum of divisors of 24.

Sol: $24 = 2^3 \times 3^1$

Sum of divisors $= \frac{2^4-1}{2-1} \cdot \frac{3^2-1}{3-1} =$ **60**

Ques: Find sum of even divisors of 24.

Sol: $(2+2^2+2^3)(1+3) =$ **56**




Ques: Find the no. of numbers between (1 to 100) which are divisible by either 2 or 5.

Sol: $A_2 \longrightarrow$ No.'s which are divisible by 2

$= \{2, 4, 6, \ldots\ldots\ldots 100\}$

$A_5 \longrightarrow$ No.'s which are divisible by 5

$= \{5, 10, 15, \ldots\ldots\ldots 100\}$

$n(A_2) + n(A_5) - n(A_2 \cap A_5)$

$50 + 20 - 10 =$ **60**




Sets are collection of well defined objects and are denoted by capital letters.

(1) Constituent of sets is called elements and are denoted by small letters like $x, y, z, a_1, a_2, \ldots\ldots\ldots$

If $x$ belong to set $A \Rightarrow x \in A$

Algebra of sets :-

Union of 2 sets:

$A = \{1, 2, 3\}$ $B = \{2, 5, 7\}$

$A \cup B = \{x : x \in A \text{ or } x \in B\}$

$A \cup B$ = Collection of elements which are in A or in B

② Intersection of 2 sets:

$A \cap B = \{x : x \in A \text{ and } x \in B\}$

$A \cap B$ = Collection of elements which are in set A and as well as in Set B

③ Compliment of a set:

$\bar{A} = \{x : x \in \Omega \text{ but } x \notin A\}$ $\cup = \Omega$ = Universal set

$A = \{1, 2, 3\}$ $\cup = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$

$\bar{A} = \{4, 5, 6, 7, 8, 9\}$

④ Substraction of 2 sets:

$A - B = \{x : x \in A \text{ but } x \notin B\}$

$A = \{1, 2, 3\}$ $B = \{2, 5, 7\}$

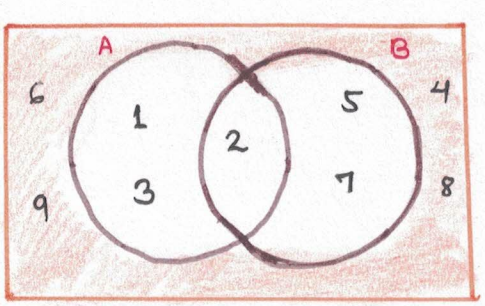
$A - B = \{1, 3\}$ $B - A = \{5, 7\}$

## VENN DIAGRAM

Geometrical representation of sets.

Universal set = $\cup = \Omega =$  Set A, B = 

① $A \cup B$




$n(A)$ = no. of elements in set A

$n(A) = 3$ $n(B) = 3$

$n(A \cup B) = 5$

$n(A \cap B) = 1$

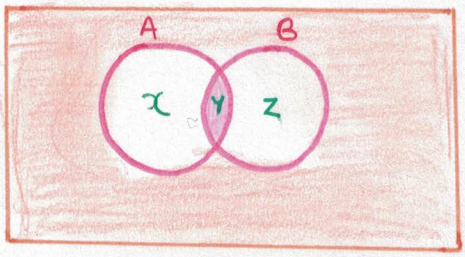
$n(A - B) = 2$

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

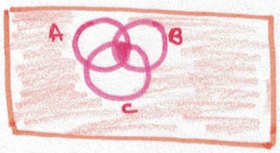
$n(A \cup B) = x + y + z$

$n(A) = x + y$

$n(B) = y + z$




(2) $n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(C \cap A) + n(A \cap B \cap C)$




(3) $n(A \cup B \cup C \cup D) = \sum n(A) - \sum n(A \cap B) + \sum n(A \cap B \cap C) - n(A \cap B \cap C \cap D)$

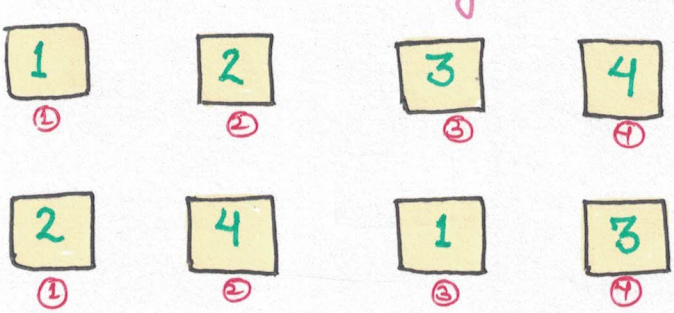
(4) $n(A_1 \cup A_2 \cdots \cup A_\lambda)$

$$= \sum_{i=1}^{n} n(A_i) - \sum\sum_{1 \le i < j \le n} n(A_i \cap A_j) + \sum\sum\sum_{1 \le i < j < k \le n} n(A_i \cap A_j \cap A_k)$$

$$\cdots\cdots (-1)^{r-1} \cdot n(A_1 \cdot A_2 \cdots A_r)$$

## DERANGEMENT

Arrangement in which none of the object is at correct (right) place. i.e. all the objects are wrongly placed.




No. of ways of derangement =

$$n!\left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots\cdots (-1)^n \frac{1}{n!}\right]$$

Where, n = no. of objects

**Ques:** Find the no. of ways of 4 letter can be put to 4 addressed envelope so that none of the letter go to right envelope.

**Sol:**

$$4!\left[1-\frac{1}{1!}+\frac{1}{2!}-\frac{1}{3!}+\frac{1}{4!}\right]$$

$$=4!\left[\frac{4!-4!+3\times4-4+1}{4!}\right]$$

$$=12-3$$

$$=9$$ ANS.

**Ques:** Find the no. of ways in which atleast one goes to correct envelope.

**Sol:**

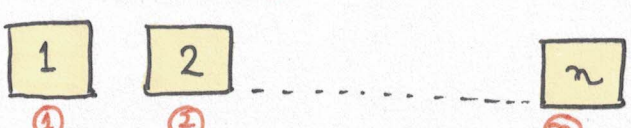
Total Ways = $4!$ = 24

Wrong entries = 9

Required Ways = 24 − 9 = **15**

**PROOF :-**

N objects and n boxes




$A_1$ = Arrangement in which ball (1) is placed correct

$A_2$ = Arrangement in which ball (2) is placed correct

$A_3$ = Arrangement in which ball (3) is placed correct

$A_i$ = Arrangement in which ball (i) is placed correct

Atleast one ball goes to correct box = Total − No. of derangement

⇒ No. of derangement = Total − Atleast one ball goes to correct box

➡ Total- $n(A_1 \cup A_2 \cup A_3 \dots A_n)$

➡ $n! - \left[\sum n(A_i) - \sum_{1 \le i < j \le n} \sum n(A_i \cap A_j) + \sum_{1 \le i < j < k \le n} \sum \sum (A_i \cap A_j \cap A_k)\right.$

$\left. + \dots\dots (-1)^n n(A_1 \cap A_2 \dots A_n)\right]$

➡ $n! - [n \times (n-1)! - {}^nC_2 (n-2)! + {}^nC_3 (n-3)! - {}^nC_4 (n-4)!$

$+ \dots\dots (-1)^{n-1} {}^nC_n \times 1]$

➡ $n! - \frac{n!}{1!} + \frac{n!}{2!} - \frac{n!}{3!} + \dots\dots \frac{(-1)^n n!}{n!}$

No. of derangement $= n!\left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \dots\dots \frac{(-1)^n}{n!}\right]$